

Lecture 1

Introduction to Digital VLSI System-on-Chip (SoC) Design

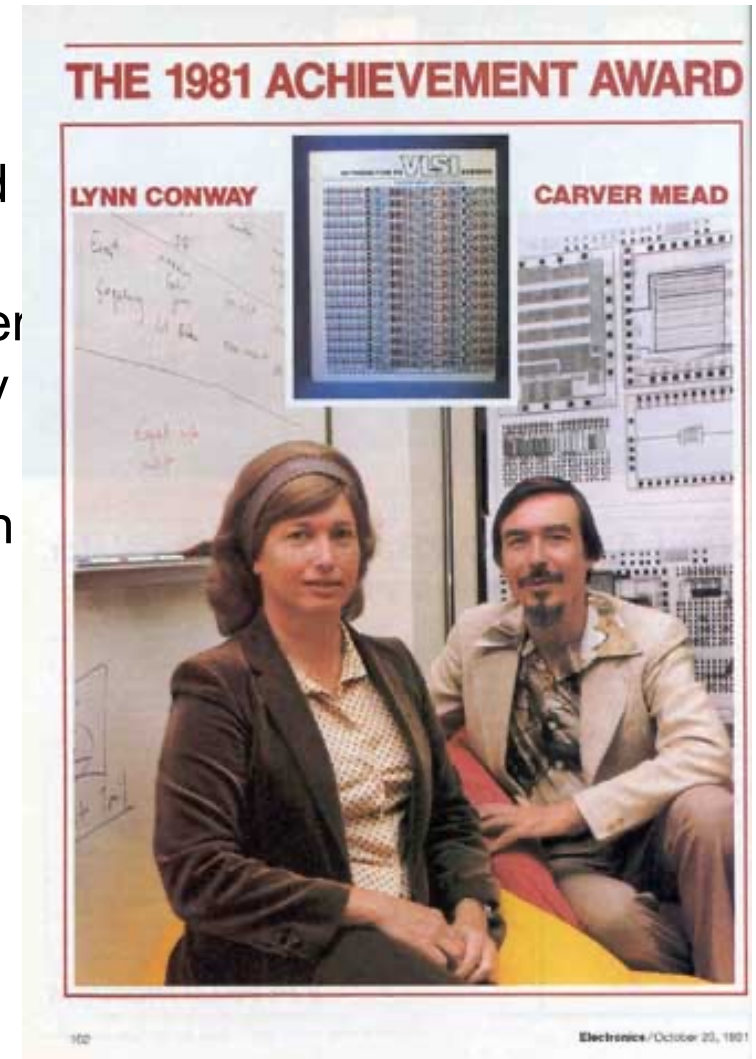
Prof Peter YK Cheung

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URL: www.ee.ic.ac.uk/pcheung/teaching/EE4-VLSI/
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Why this module? (1)

- ◆ Mead & Conway published a book in 1979 – showed students can design their own chips
- ◆ Between 1992 – 2010, I taught an elective called “Introduction to Digital IC Design”
- ◆ From 2010 onwards, Tim Constandinou took over to teach “Full-Custom IC Design” which is mainly analogue circuits
- ◆ VLSI chips are now much larger and denser than in 2011
- ◆ Design methodology is now very different

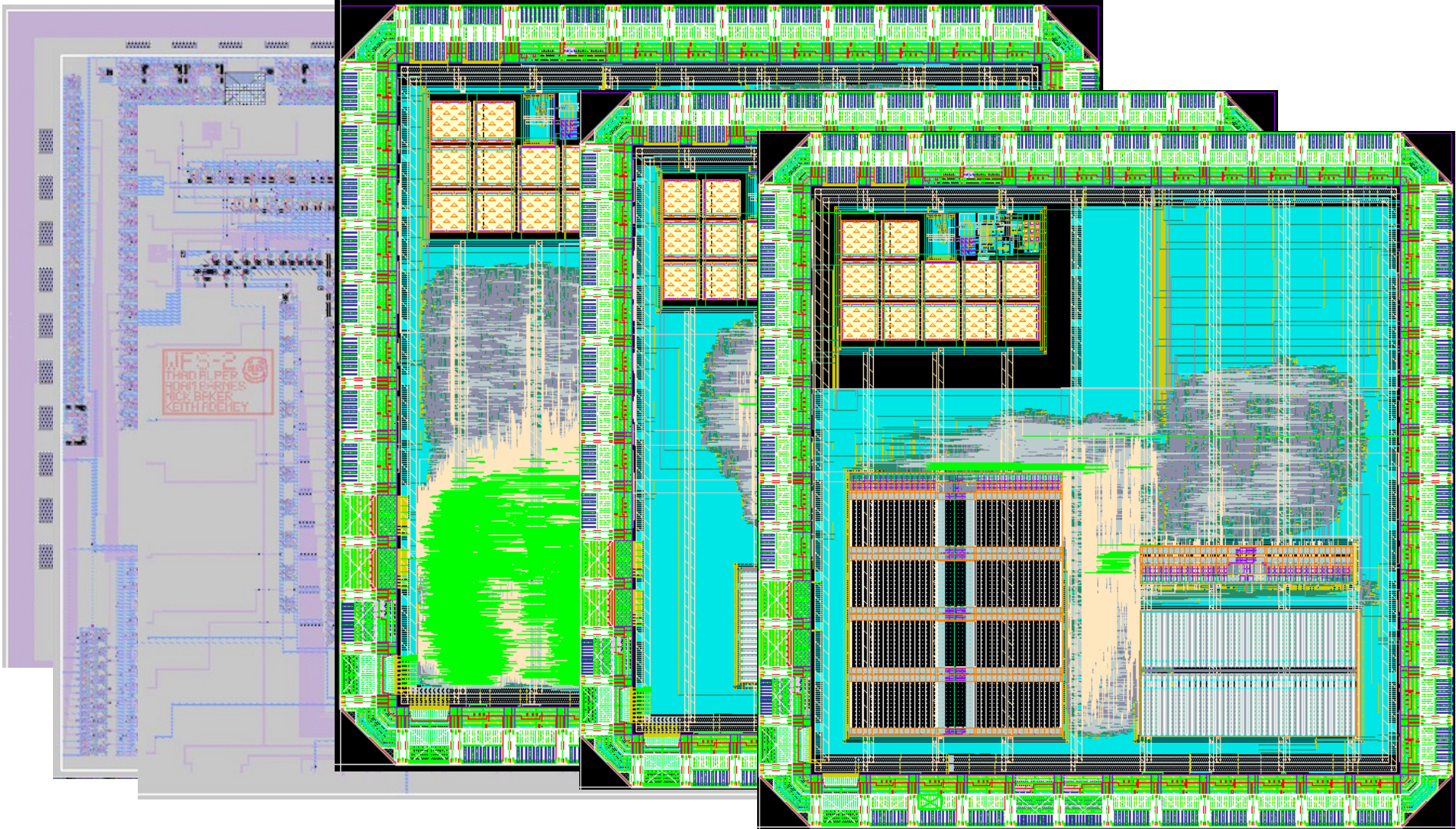


Why this module? (2)

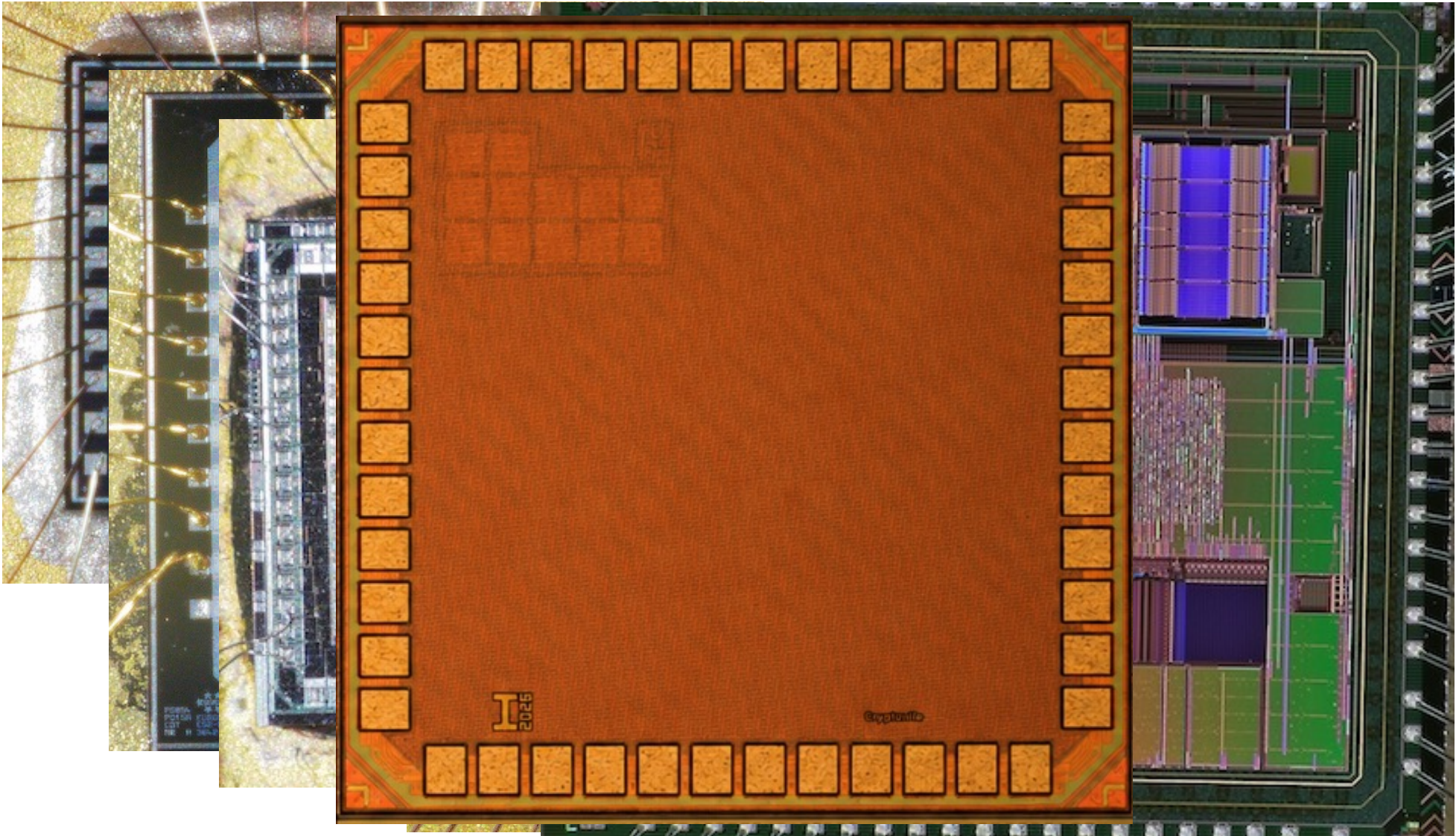
- ◆ Used to make students chips in early days
- ◆ Stopped due to high cost and chips took too long to come back
- ◆ Most were NOT tested

- ◆ Apple Inc. approached us to deliver a digital VLSI module – from specification to silicon tapeout
- ◆ They provide a grant to make chips for students again
- ◆ They help to deliver some of the industrial-related teaching

Examples of students' chip layout



Examples of fabricated students' designs



Teaching Team



Peter Cheung
(Module lead)



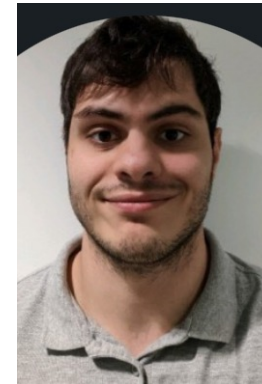
Sam Coward
(Deputy lead)



Peter Liu
(GTA)



Mayank Manas
(GTA)



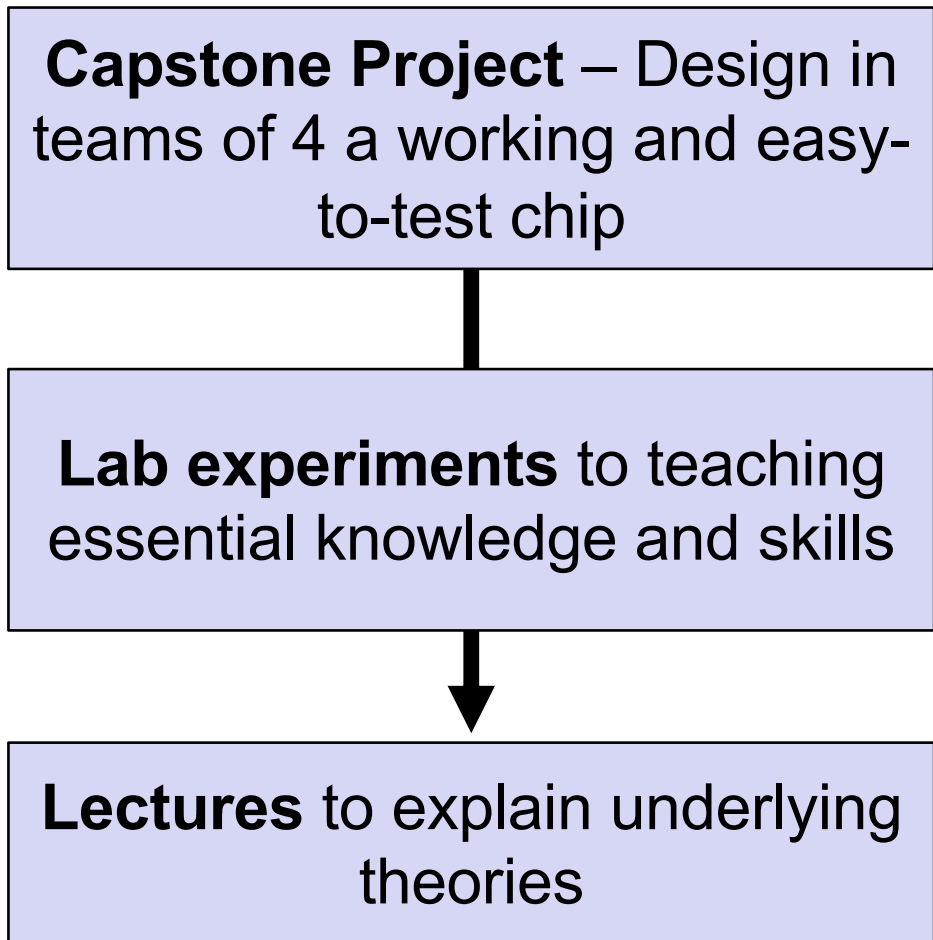
Bardia Zadeh
(GTA)

Learning Outcomes

By the end of this module, students will be able to:

- ◆ Describe the **digital VLSI design flow**
- ◆ Explain the **trade-offs between the different IC design approaches**
- ◆ Understand key concepts in the **fabrication process** and the role of **process design kits (PDKs)**
- ◆ Produce **cell-based circuit** from a system-level specification
- ◆ **Design and simulate** a moderately complex digital IC
- ◆ Understand circuits from **layout**
- ◆ Perform **logical and physical verification** of the design using DRC, LVS, and other tools
- ◆ **Design for** easy post-manufacturing **testing**
- ◆ Generate a complete **tapeout deck** ready **for silicon fabrication**.
- ◆ Test that the **fabricated chip is working**

Course Design - Back-to-front



Assessment & Textbook

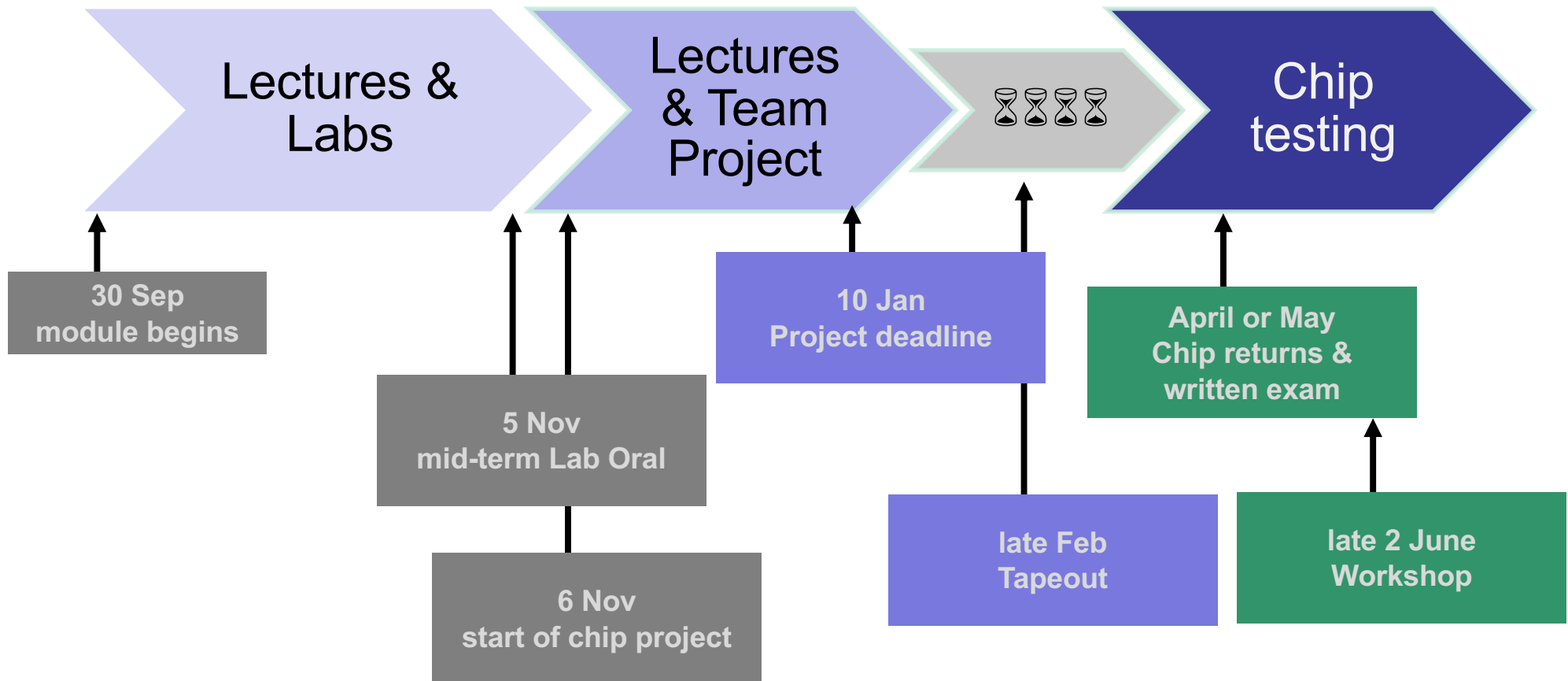
- ◆ Mid-term Lab Oral (Labs 1 to 4) (10%) – 5 November 2026
- ◆ Team Chip Design Project (50%)
- ◆ Written examination in Summer Term (30%)
- ◆ Summer Term chip test & team presentation (10%)

- ◆ Textbook:
“CMOS VLSI Design: A Circuits and Systems Perspective”, 4th Edition, ET Weste & David Harris, Pearson, 2015

Preliminary Course schedule (2026)

Week	Lectures	Labs
1	1. Introduction and Overview 2. Design methodologies and design flow	Lab 1: Quick start with Synopsys
2	3. Fabrication Process 4. CMOS Inverter & Layout Design	Lab 2: Understanding Layout (Part 1)
3	5. Full-custom combinational circuits	Lab 3: Understanding Layout (Part 2)
4	6. Full-custom sequential circuits	Lab 4: Memory Compiler and self-test
5	7. Chip Design Project Briefing	Lab 5: Verification and Test
6	8. Array Subsystems	Mid-term Lab Oral Examination
7	9. Chip Testing 10. Design for Testability	Team Project Session 1
8	11. Tapeout Process & Chip Bring-up	Team Project Session 2
9	12. VLSI Delays 13. VLSI Interconnects	Team Project Session 3
10	14. Reliability & Variations in VLSI 15. Packaging, Power & Clock Distribution	Team Project Session 4
11	Team Project due on 10 th Jan 2027 at 23.59 GMT	Team Project Session 5

Key Dates

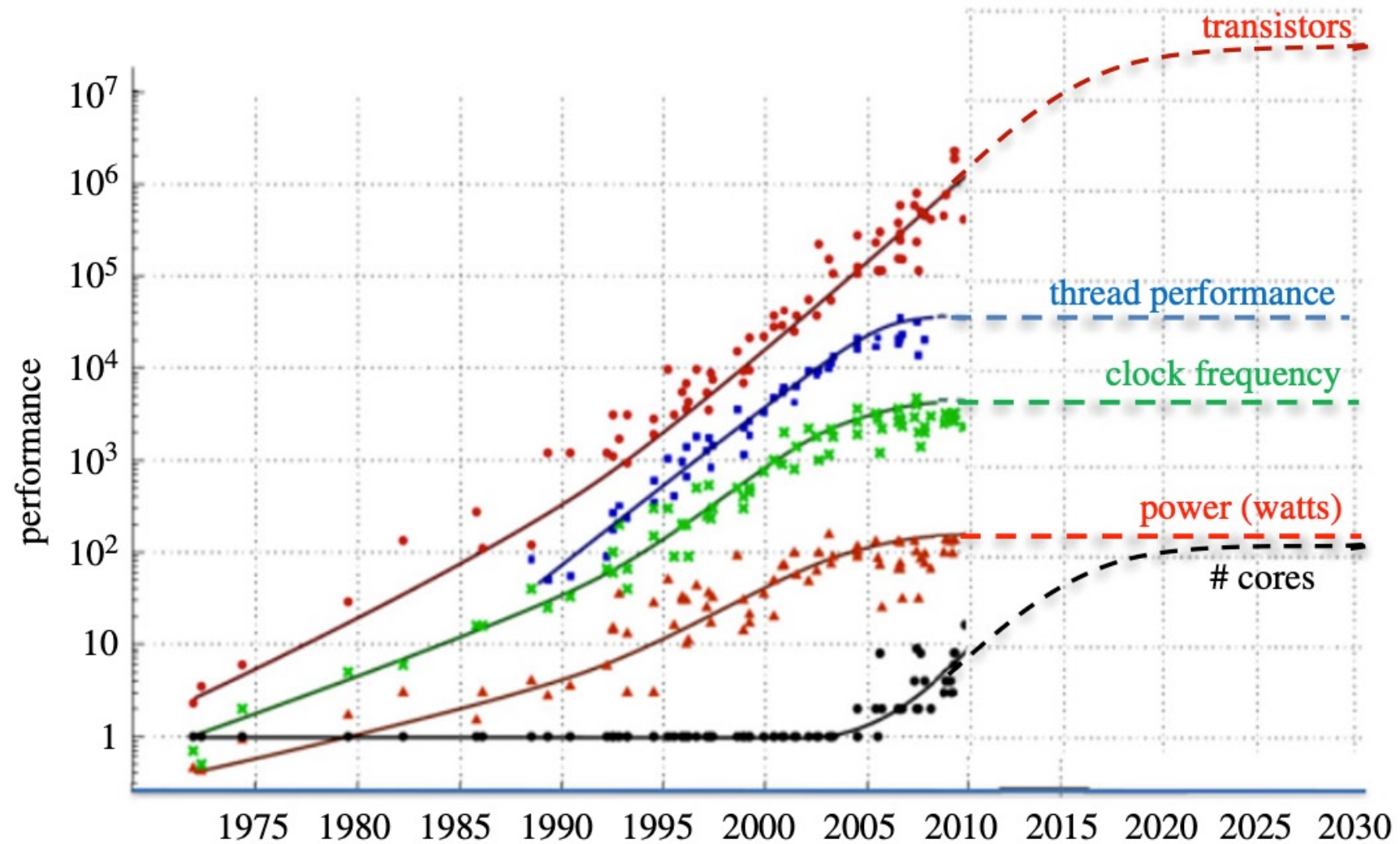


What you **MUST** do before tomorrow's Lab

- ◆ Complete and sign the two TSMC NDA forms
- ◆ Return forms to me immediately during the lab session
- ◆ Form teams of four
- ◆ Complete the team partner declaration survey

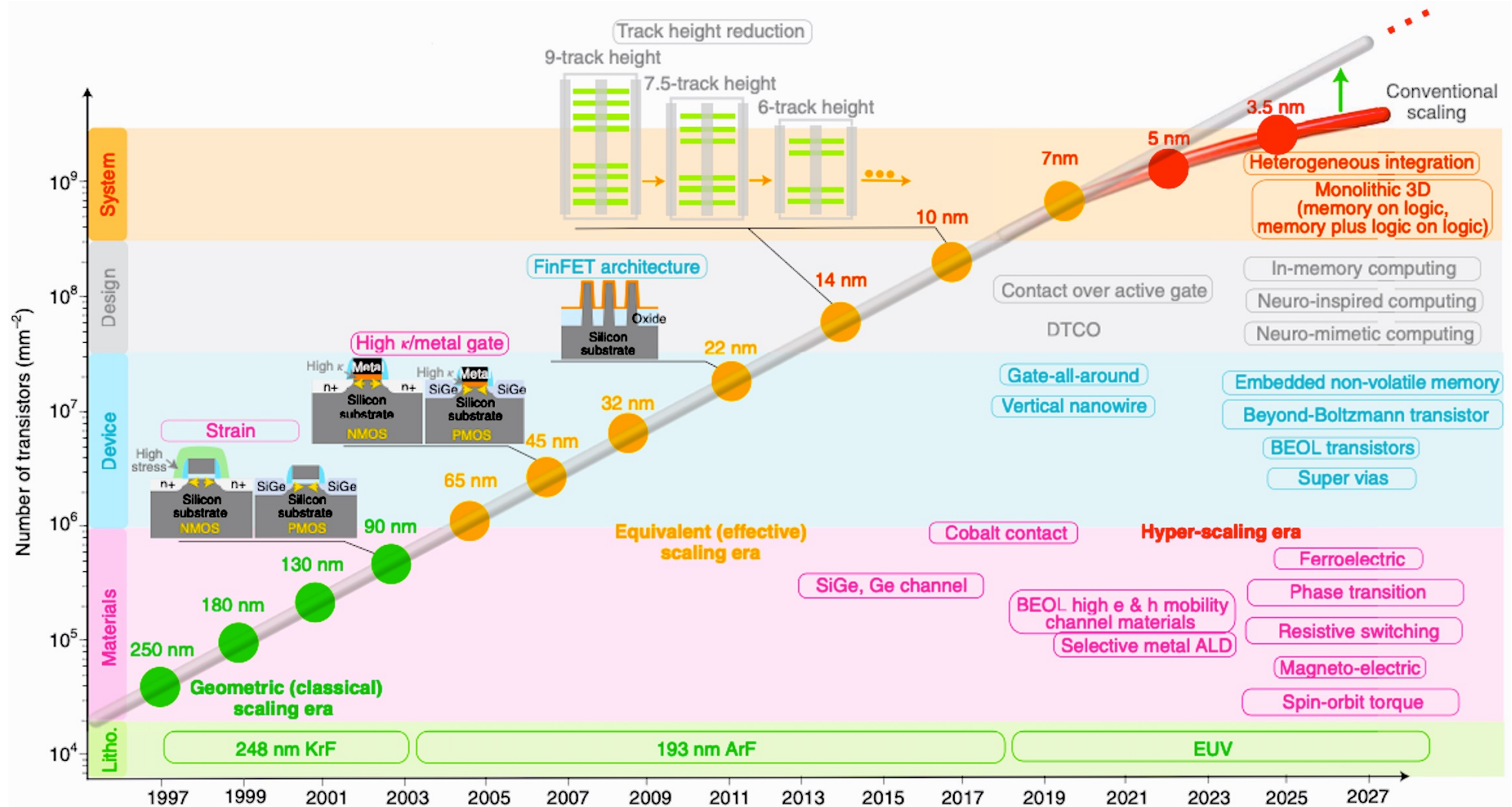
- ◆ DON'T FORGET – you cannot back out from this module!!!!
- ◆ DON'T FORGET – you **MUST** abide by the NDA agreement

The end of Moore's Law?



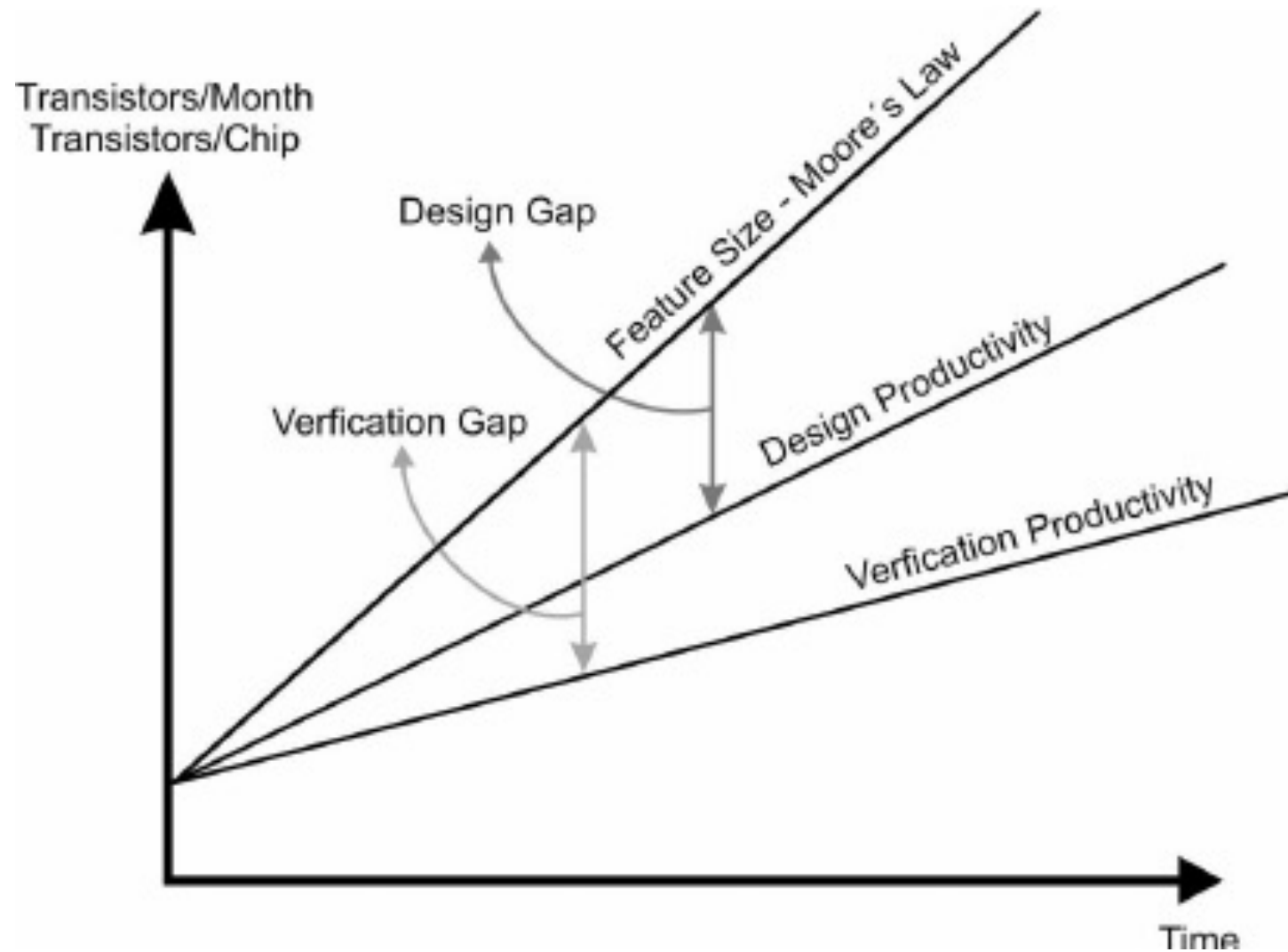
Source: John Shalf (2020), "The future of computing beyond Moore's Law"

Process Technology Scaling



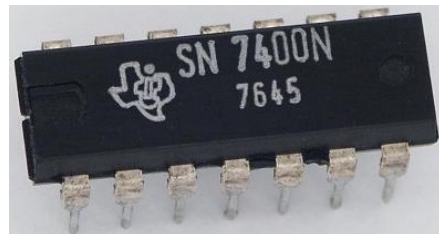
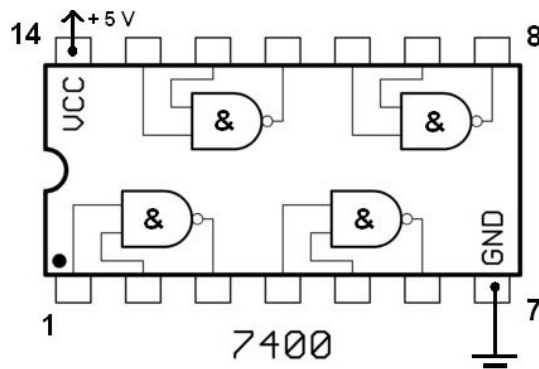
Source: Salahuddin, Ni and Datta, Nature electronics 2018

Tackling the Complexity Challenge



Source: Oswaldo Cadenas (2006)

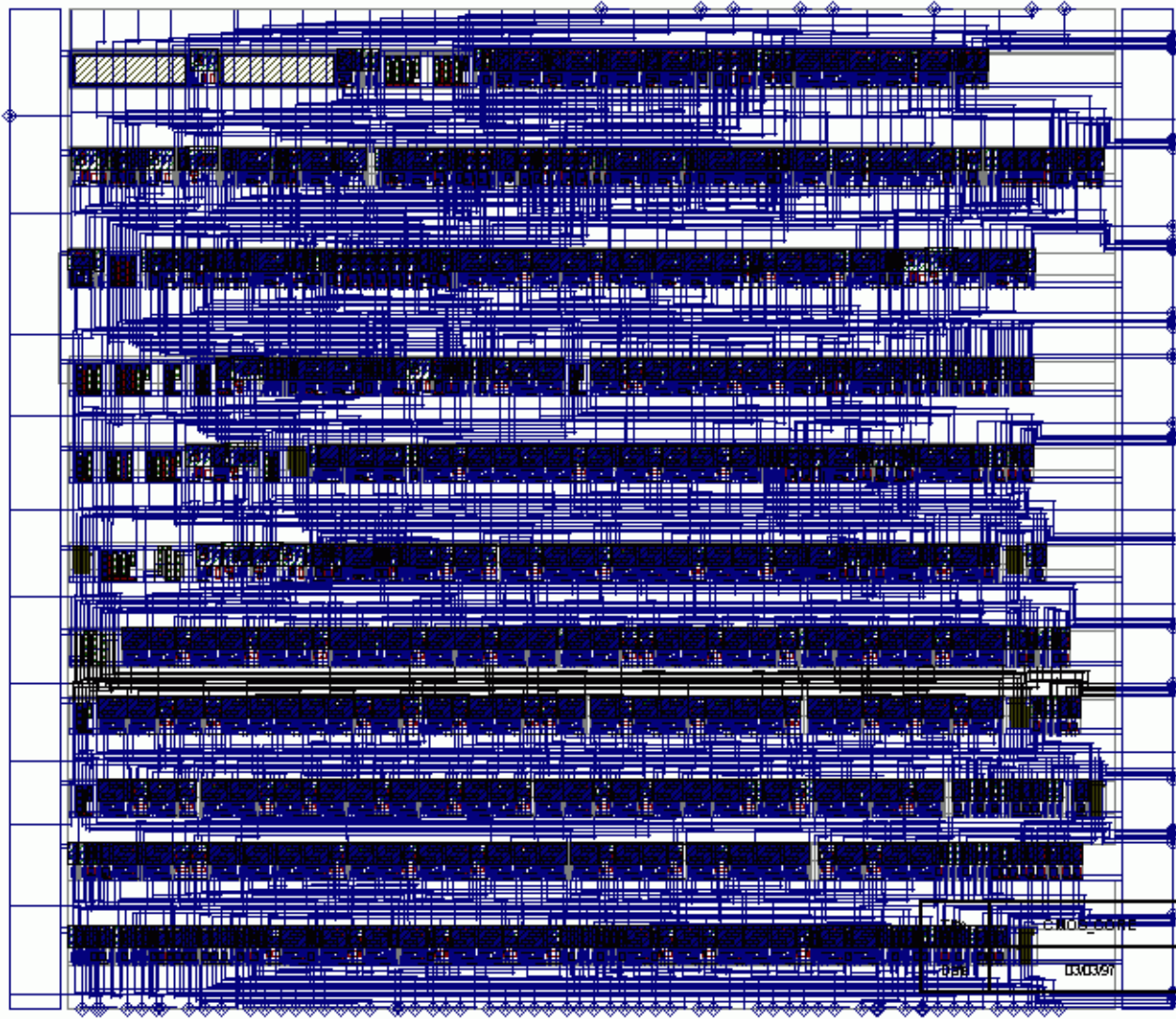
Old ways of implementing digital circuits



- ◆ Discrete logic – based on gates or small packages containing small digital building blocks (at most a 1-bit adder)
- ◆ De Morgan's theorem – theoretically we only need 2-input NAND or NOR gates to build anything
- ◆ Tedious, expensive, slow, prone to wiring errors



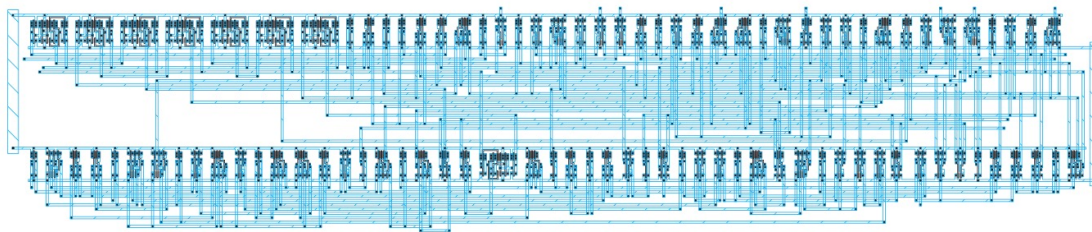
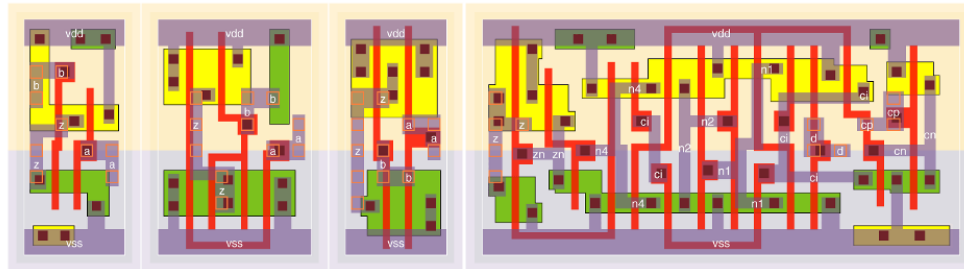
Early integrated circuits based on gate arrays



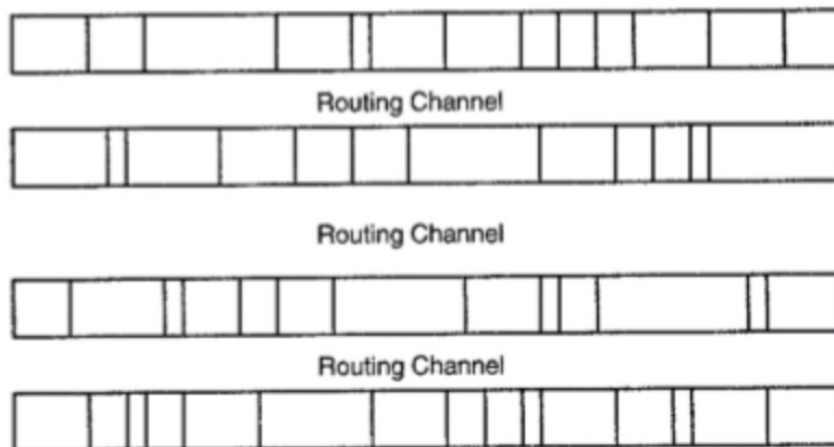
- ◆ Rows of gates – often identical in structure
- ◆ Connected to form customer specific circuits
- ◆ Can be full-custom (i.e. completely fabricated from scratch for a given design)
- ◆ Can be semi-custom (i.e. customisation on the metal layers only)
- ◆ Once made, design is fixed

Source: LSI Logic

Standard cell-based design (1)



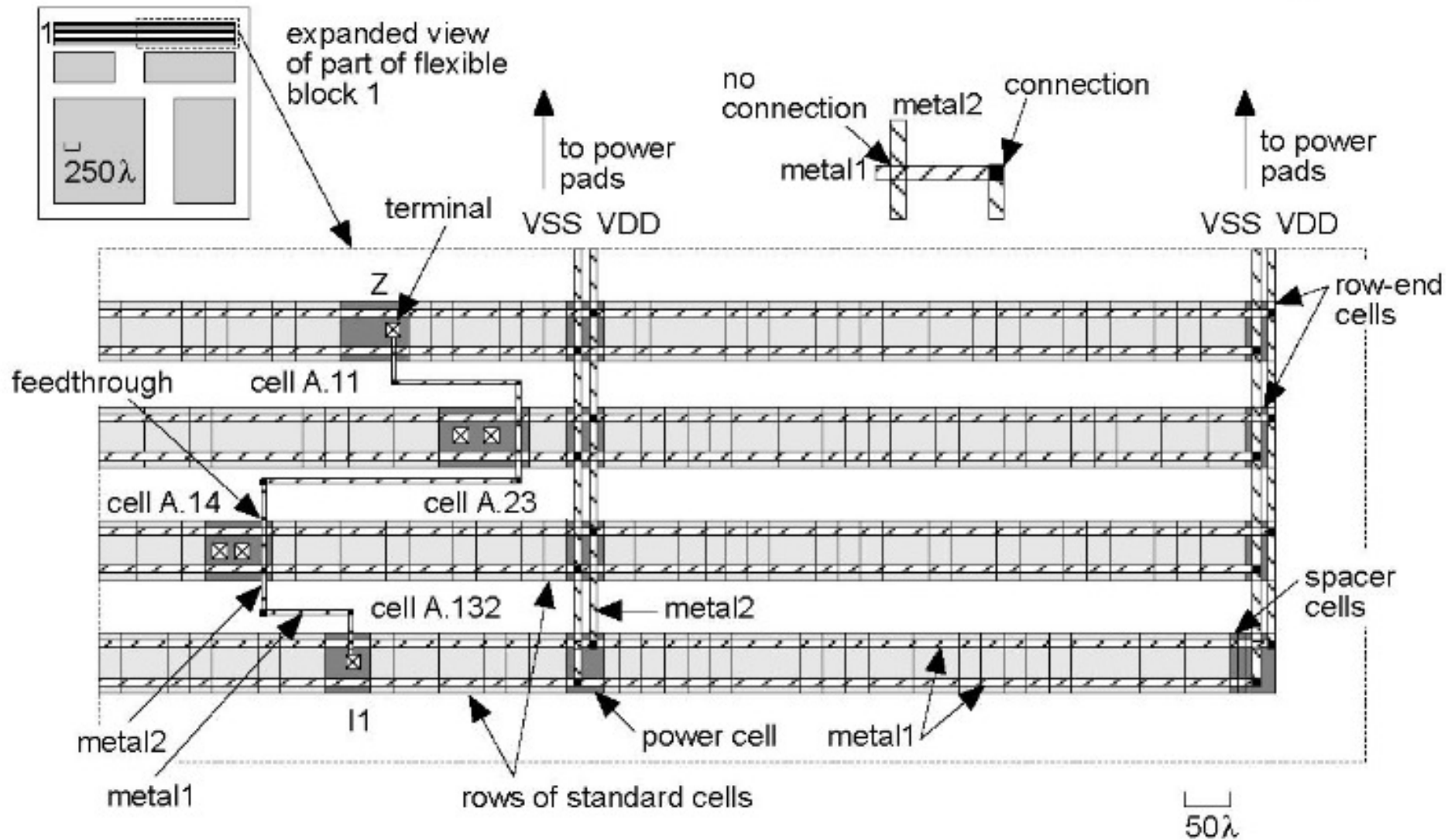
Standard-cell Row



- ◆ Based on a library of logic gates and flip-flops
- ◆ Connected to form customer specific circuits
- ◆ Pitch matched – same height on all cells
- ◆ Any gaps are filled with filler cells (bridging VDD and GND)
- ◆ Bypass cells

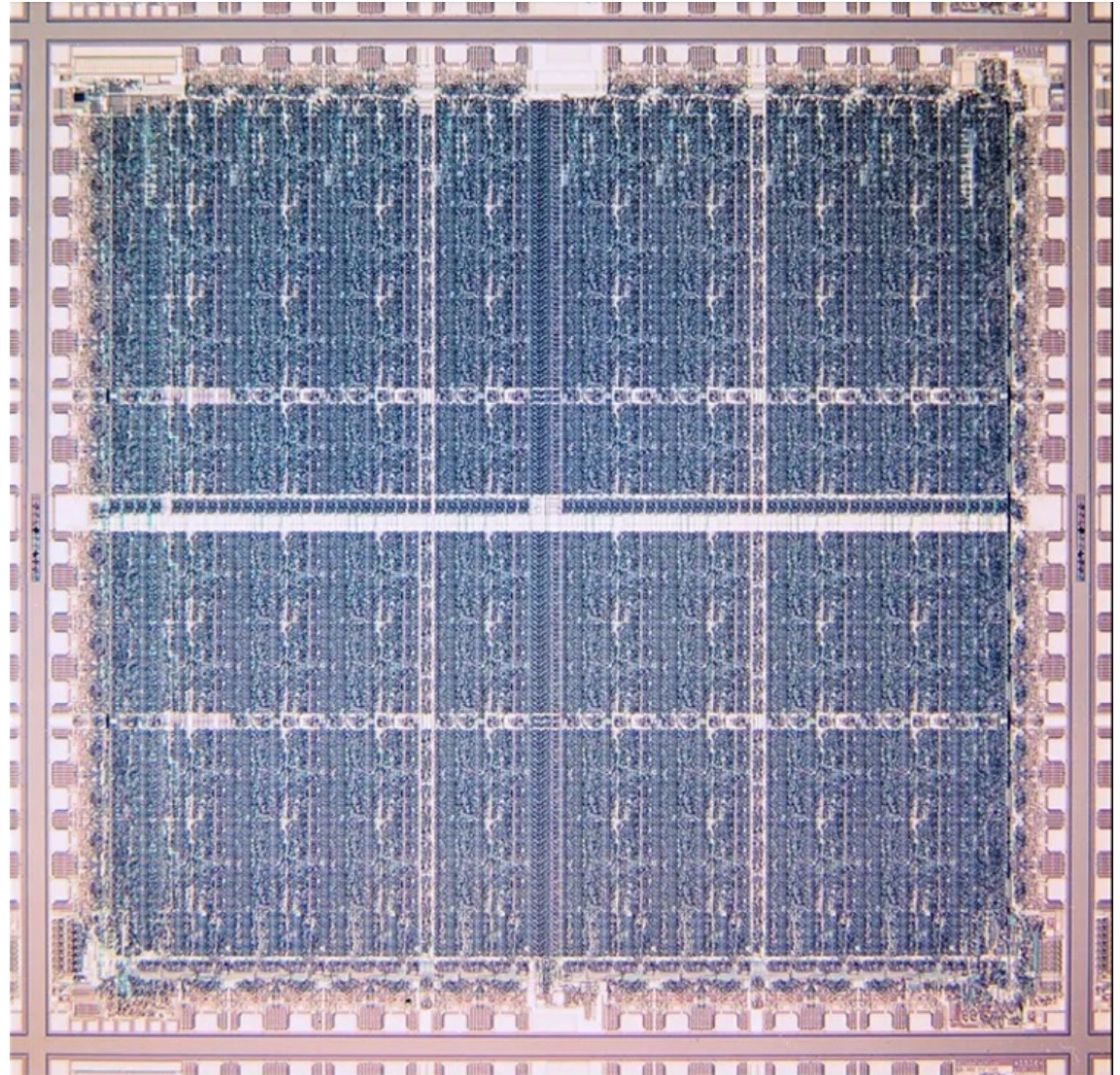
Source: VLSI Technology

Standard cell-based design (2)



FPGAs

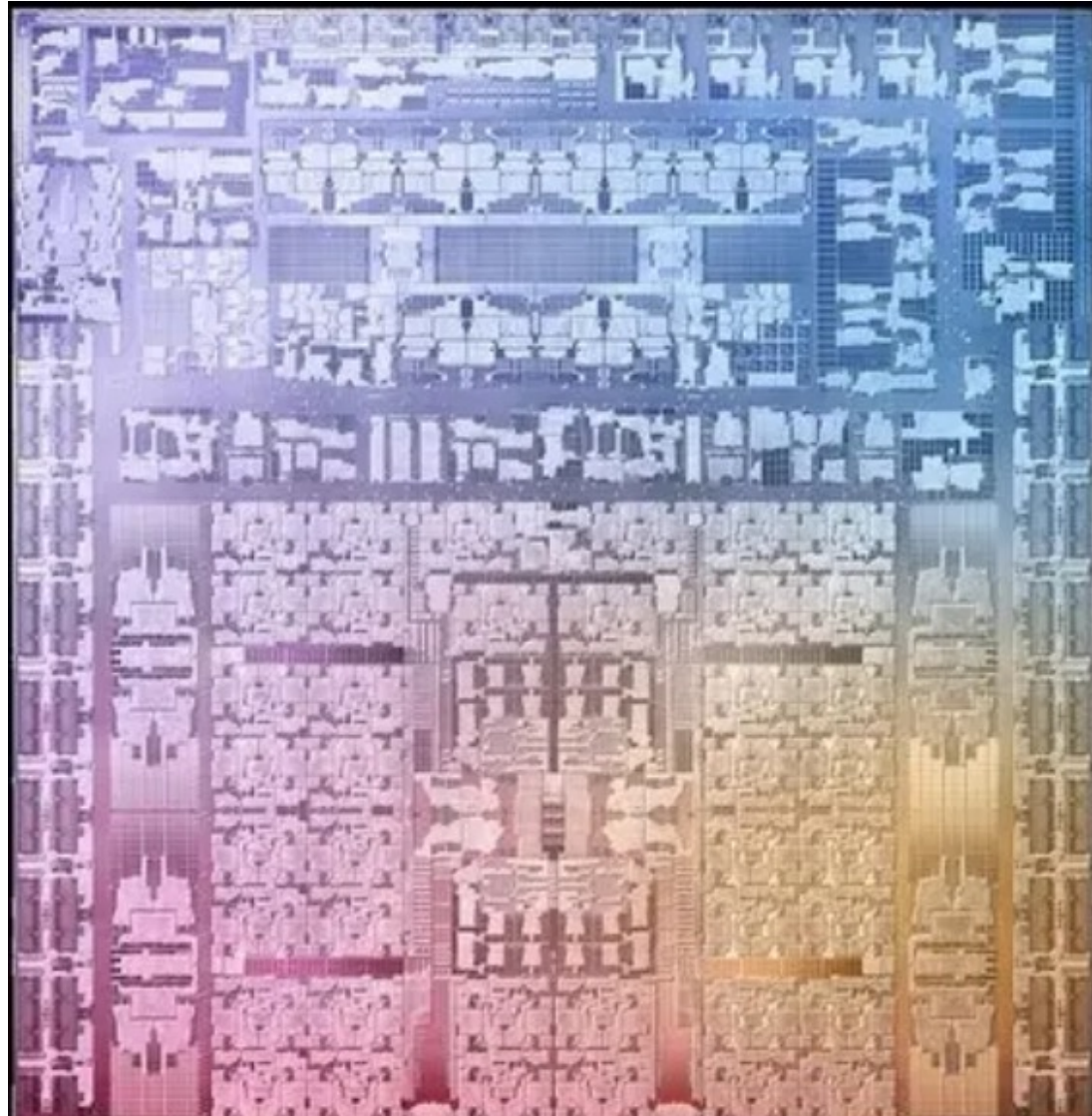
- ◆ XC2064 die (1985)
- ◆ 2um technology
- ◆ 64 CLBs (~64k trans)
- ◆ Latest AMD/Altera FPGAs have 30-40 billion trans.
- ◆ Area cost around 10x to 30x higher than ASIC
- ◆ Power may be 10x higher
- ◆ Suitable for <~100k units



Source: Xilinx/IEEE Spectrum 2017

Modern digital design – full custom IC

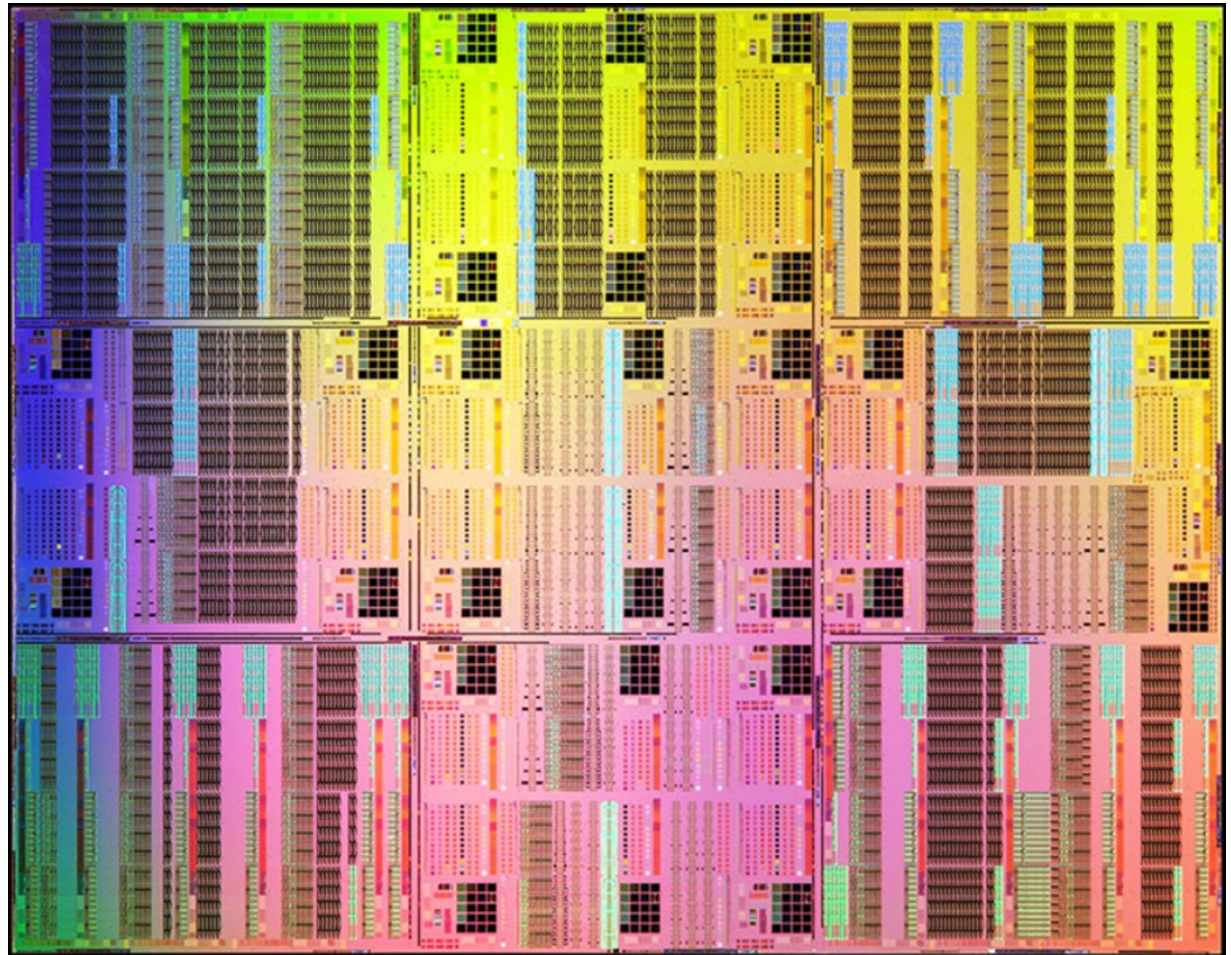
- ◆ Apple M3 Max
- ◆ 92 billion trans.
- ◆ 3 nm TSMC technology
- ◆ Very expensive to design
- ◆ Very expensive to manufacture
- ◆ Not viable unless $>\sim 100k$ units



Source: Tom's Hardware 2023

Latest in 2026 – 100 billion transistor chip

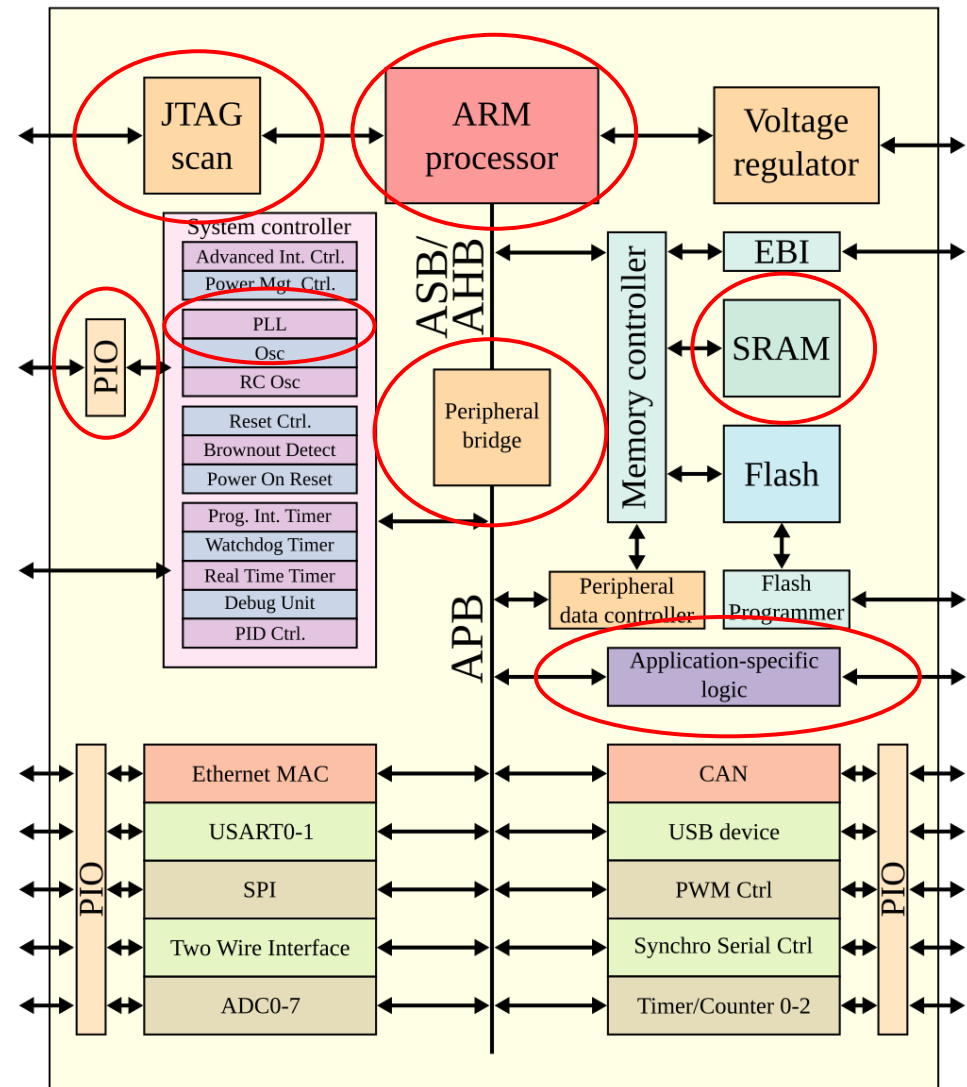
- ◆ IBM announced in June 2026
- ◆ 100 billion transistor chip
- ◆ ~ 1cm x 1cm area
- ◆ 0.7nm technology
- ◆ “Nanostack” 3D architecture
- ◆ 40% SRAM scaling



Source: IBM, June 25, 2026

A Typical System-on-Chip (SoC) Design

- ◆ Here is a typical SoC
- ◆ You chip could have:
 - ARM or RISC-V processor
 - Memory
 - PIO
 - Application specific logic
 - Bus bridge
 - PLL
 - JTAG scan
 -



Source: wikipedia

Comparison of VLSI Implementations

	Full Custom	Standard Cell	Gate Array	FPGA
Performance	V. High	High	Medium	Low
Density	V. High	High	Medium	Low
Development Cost	V. High	Medium	Medium	Low
Part cost	V. Low	Medium	Medium	High
Design Time	V. Long	Medium	Short	V. Short
Testability	Difficult	Less difficult	Easy	V. Easy
Application volume	High volume	Medium volume	Low	Low to V. Low

Further Reading

- ◆ You may want to read the textbook Weste & Harris, sections 1.1, 1.2, 1.6, 1.8, 1.9.
- ◆ This document outlines the roadmap for VLSI in the future:
International Roadmap for Devices and Systems, 2024 (Executive Summary)

https://irds.ieee.org/images/files/pdf/2024/2024IRDS_ES.pdf